A rational-input specialization of Kelner's sparse linear solver

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Abstract

Kelner's recent identity-modulus lifting method gives a deterministic route to the quadratic bit-work scale for sparse linear systems. We specialize this method to the short rational inputs in AIM problem 044. For each fixed positive integer c, a nonsingular rational $n \times n$ matrix with m nonzeros, condition number at most n^c, and input numerators and denominators of $O_c(\log(2n))$ bits can be solved to relative Euclidean solution error at most $n^{-c}/10$ in $\widetilde{O}_c(nm)$ bit operations. Individual rational rounding avoids a common denominator over the input. The proof gives explicit scales, short residual bounds, reconstruction heights, and a direct forward-error estimate. An accompanying implementation uses FLINT for fast bounded rational reconstruction and includes exact finite-case checks.

1 The promised-input problem

The distinction between arithmetic operations and bit operations matters for sparse linear systems: even an algorithm with few matrix–vector products may operate on increasingly long numbers. The finite-precision question appears as Problem 2.10 of the Simons workshop collection [2]. AIM problem 044 [1] fixes a polynomial condition bound and short rational input, and asks for $\widetilde{O}(nm)$ bit operations at inverse-polynomial relative solution accuracy.

Kelner [3] introduced a deterministic solver using a perturbation whose scaled matrix is the identity modulo a chosen integer. Its lifting steps need only sparse products on short integers; fast rational reconstruction recovers the solution of the perturbed system. That method and its fast reconstruction theorem are the basis of this note. Our purpose is to make its consequence for the exact rational-input and forward-error formulation of problem 044 explicit. The algorithmic mechanism is due to Kelner.

All vector norms without a subscript are Euclidean, and matrix norms with subscript 2 are induced Euclidean norms. We use $\kappa_2(A) = \|A\|_2 \left\|A^{-1}\right\|_2$. A rational number is represented by a signed numerator and a positive denominator. Bit work includes arithmetic, input and output, and sparse indices. The notation $\widetilde{O}_c$ hides fixed powers of $\log(2n)$, with constants allowed to depend on the fixed integer c.

Theorem 1. *For every fixed positive integer c there is a deterministic algorithm with the following guarantee. Let $n \geq 1$, let $A \in \mathbb{Q}^{n\times n}$ be nonsingular and given by its sparse nonzero list, put $m = \operatorname{nnz}(A)$, and let $b \in \mathbb{Q}^n \setminus \{0\}$. Suppose*

$$\kappa_2(A) \leq n^c, \qquad B = c\lceil \log_2(2n) \rceil,$$

and every input numerator and denominator has at most B bits. In $\widetilde{O}_c(nm)$ bit operations the algorithm returns a rational vector $\widehat{x}$ satisfying

$$\left\|\widehat{x} - A^{-1}b\right\|_2 \le \frac{n^{-c}}{10}\left\|A^{-1}b\right\|_2. \tag{1}$$

The probability of success is one, which meets the randomized success requirement of problem 044. Symmetry, positive definiteness, and graph structure are not required. Nonsingularity and the condition bound are input promises; the algorithm need not certify them.

2 An exact integer solve by lifting

We first describe the integer part of the construction. This is Kelner's identity-modulus method, with convenient explicit bounds. For an integer matrix $C \in \mathbb{Z}^{n\times n}$ and $d \in \mathbb{Z}^n$, set

$$T = \max\{1, \max_{i,k}|C_{ik}|, \max_i |d_i|\}.$$

Choose a power of two $R = 2^s \ge 2nT$ and put $M = I + RC$. The determinant expansion gives $\det M \equiv 1 \pmod R$. Thus M is nonsingular, and its determinant is coprime to every power of R. In particular this construction does not require C to be nonsingular.

To solve $Mz = d$, fix $L = 6n + 2$, start with $r_0 = d$, and compute, for $j = 0, \dots, L-1$,

$$u_j = r_j \bmod R \in \{0, \dots, R-1\}^n, \qquad r_{j+1} = \frac{r_j - u_j}{R} - Cu_j. \tag{2}$$

Remainders are taken coordinatewise and are nonnegative even for negative residuals. The divisions in (2) are exact. The digits encode the modular solution; they are stored during the iteration and assembled only after the last step.

Lemma 1. *Let $P = R^L$ and $v = \sum_{j=0}^{L-1} R^j u_j$. Then $0 \le v_i < P$ and $Mv \equiv d \pmod P$. Each residual coordinate has absolute value at most R^2, and all arithmetic operands used in the sparse products have $O(s)$ bits. The lifting and assembly cost $O(n(\mathrm{nnz}(C) + n)\,\mathrm{poly}(s + \log(2n)))$ bit operations.*

Proof. Equation (2) gives $Mu_j = r_j - Rr_{j+1}$. Multiplication by R^j and summation telescope to

$$Mv = d - R^L r_L. \tag{3}$$

The digit bounds give the asserted range of v.

For a vector, $\|\cdot\|_\infty$ denotes its largest absolute coordinate; for a matrix it denotes the largest absolute row sum. Put $W = \max\{\|d\|_\infty, \|M\|_\infty\}$. If $\|r_j\|_\infty \le W$, then

$$\|r_{j+1}\|_\infty \le \frac{W + (R-1)\|M\|_\infty}{R} \le W.$$

This starts at $r_0 = d$. Since $nT \le R/2$,

$$\|M\|_\infty \le 1 + nRT \le 1 + R^2/2 \le R^2, \qquad \|d\|_\infty \le T \le R^2.$$

Every partial sum in a row of Cu_j is bounded in absolute value by $nT(R-1) \le R^2/2$. Residuals, digits, products, sums, and the exact division therefore involve $O(s)$ bits. Each of the $O(n)$ rounds costs $O(\mathrm{nnz}(C) + n)$ short-integer operations, with the sparse indices charged. Elementary arithmetic suffices at these widths. As R is a power of two, concatenating the s-bit digits once per coordinate constructs v in $O(nLs)$ bit operations. Repeatedly updating long prefixes is unnecessary. □

The last step uses bounded rational reconstruction. We state the precise external result because its arithmetic cost is essential.

Proposition 1 (Kelner, Theorem 2.3)**.** *Let P, v, N, D be integers with $P \geq 2$, $0 \leq v < P$, $N, D \geq 1$, and $P > 2ND$. There is at most one reduced fraction a/q satisfying*

$$|a| \leq N, \quad 1 \leq q \leq D, \quad \gcd(q, P) = 1, \quad a \equiv vq \pmod{P}.$$

If it exists, it can be recovered deterministically in $O(\mathsf{M}(k) \log k)$ bit operations, where $k = \lceil \log_2(P+1) \rceil$ and $\mathsf{M}(k) = \widetilde{O}(k)$ is the bit cost of fast integer multiplication.

This is Theorem 2.3 on page 7 of the pinned version [3]; its modulus need not be prime. The fast reconstruction algorithm is part of this statement. Ordinary scalar Euclidean reconstruction cannot replace it in the bit-work analysis.

Lemma 2. *With M, d, R, L, P, v as above, set $H = R^{3n}$. Applying Proposition 1 to each v_i with $N = D = H$ recovers exactly $z = M^{-1}d$. All coordinates together cost $\widetilde{O}(n^2 s)$ bit operations, including parameter construction and output writing.*

Proof. Write $z_i = a_i/q_i$ in lowest terms, with $q_i > 0$. By Cramer's rule, $z_i = \det(M_i)/\det M$, where M_i replaces column i by d. Hence q_i divides $|\det M|$ and is coprime to P.

All entries in M and d have absolute value at most $RT + 1 \leq R^2$. Hadamard's determinant bound gives

$$|\det M|, \ |\det(M_i)| \leq n^{n/2} R^{2n} \leq R^{5n/2} \leq H,$$

where $n \leq R$. For completeness, orthogonalizing the columns preserves the determinant and does not increase their lengths, so its absolute value is bounded by the product of the original column lengths. A dependent column gives determinant zero. Reduction of the Cramer fractions therefore preserves $|a_i| \leq H$ and $q_i \leq H$.

Since M is invertible modulo P, (3) implies $a_i \equiv v_i q_i \pmod{P}$. Also

$$P = R^{6n+2} \geq 4R^{6n} > 2H^2.$$

All hypotheses of Proposition 1 are now satisfied, including existence and coprimality. The modulus has $O(ns)$ bits, giving $\widetilde{O}(ns)$ work per coordinate. The parameters P, H are binary powers of two, constructed by writing their bits. Every output numerator and denominator has $O(ns)$ bits, and their total writing cost is $O(n^2 s)$. □

3 Rational rounding and forward error

We now choose the integer data for Theorem 1. Put $Q = 2^B$ and $K = n^c$. Each nonzero input has absolute value at least $1/Q$ and each input has absolute value at most Q. Nonsingularity gives a nonzero matrix entry, and a nonzero column tested on a coordinate vector therefore shows $\|A\|_2 \geq 1/Q$. Consequently

$$\left\|A^{-1}\right\|_2 \leq KQ, \qquad \|b\|_2 \geq 1/Q, \qquad x = A^{-1}b \neq 0. \tag{4}$$

Every row of A is nonzero, so $m \geq n$.

To keep the perturbation small even at the largest allowed condition number, choose

$$\ell = \lceil \log_2 n \rceil, \quad h = 4 + B + (2c+1)\ell, \quad S = 2^h, \qquad S \geq 16K^2 nQ. \tag{5}$$

Round entries individually by

$$C_{ik} = \mathrm{round}(SA_{ik}), \qquad d_i = \mathrm{round}(Sb_i), \qquad \mathrm{round}(t) = \lfloor t + 1/2 \rfloor.$$

On an input p/q with $q > 0$, the rounded value is $\lfloor (2Sp+q)/(2q) \rfloor$. This requires only $O_c(\log(2n))$-bit operands. Zeros remain zero, so $\mathrm{nnz}(C) \le m$, and no common input denominator is formed. Entrywise rounding errors give

$$\|C/S - A\|_2 \le \frac{n}{2S}, \qquad \|d/S - b\|_2 \le \frac{\sqrt{n}}{2S}. \tag{6}$$

The first inequality follows by bounding the operator norm by the Frobenius norm, using Cauchy–Schwarz on each row.

For these C, d, the quantity T in Section 2 obeys $T \le SQ + 1/2$, so $\log_2 T = O_c(\log(2n))$. Choose the least power of two $R \ge 2nT$, form $M = I + RC$, and use the lifting and reconstruction of Section 2 to obtain $z = M^{-1}d$. Return

$$\widehat{x} = Rz. \tag{7}$$

The complete computation is summarized in the algorithm below.

Algorithm: sparse rational solve
Input: n, c, A, b satisfying Theorem 1.
Output: $\widehat{x} \in \mathbb{Q}^n$.

Compute B, ℓ, h, S by (5) and the definition of B.
Set $C = \mathrm{round}(SA)$ and $d = \mathrm{round}(Sb)$ entrywise.
Set $T = \max\{1, \max|C_{ik}|, \max|d_i|\}$; choose $R = 2^{\lceil \log_2(2nT) \rceil}$.
Set $L = 6n + 2$ and $r = d$.
for $j = 0, \ldots, L-1$ **do**
  $u_j = r \bmod R$; store the n radix-R digits.
  $w = Cu_j$ using the sparse nonzero list of C.
  $r = (r - u_j)/R - w$.
end for
Set $P = R^L$ and $H = R^{3n}$ as binary powers of two.
for $i = 1, \ldots, n$ **do**
  Concatenate the stored digits to form $v_i = \sum_{j=0}^{L-1} R^j (u_j)_i$.
  Recover a_i/q_i from $v_i, P, N = D = H$ by Proposition 1.
  $\widehat{x}_i = Ra_i/q_i$.
end for
return $\widehat{x}$.

Lemmas 1 and 2 give $\widetilde{O}_c(n(m+n) + n^2) = \widetilde{O}_c(nm)$ bit work for this computation, including rounding, assembly, reconstruction and output. Stored digits and sparse data, with fast reconstruction workspace, require $\widetilde{O}_c(n^2)$ bits. Diagnostics and exact reference solutions used in the tests are outside the solver. The following proof establishes its relative error and collects the work estimates.

Proof of Theorem 1. The exact working equation $Mz = d$ becomes

$$(A + E)\widehat{x} = b + f, \qquad E = C/S - A + I/(RS), \quad f = d/S - b.$$

Both perturbations must be included in the forward error. By (6) and $R \geq 2$,

$$\|E\|_2 \leq n/S, \qquad \|f\|_2 \leq n/(2S).$$

Let $\beta = \|A^{-1}\|_2 \|E\|_2$. Equations (4)–(5) yield

$$\beta \leq \frac{KnQ}{S} \leq \frac{1}{16K} \leq \frac{1}{16}.$$

For any vector w, the triangle inequality gives $\|(I + A^{-1}E)w\|_2 \geq (1-\beta)\|w\|_2$. This square matrix is injective, hence invertible, with inverse norm at most $(1-\beta)^{-1}$. Subtracting the equation $Ax = b$, and writing $e = \widehat{x} - x$, gives

$$(I + A^{-1}E)e = A^{-1}f - A^{-1}Ex.$$

As $\|b\|_2 = \|Ax\|_2 \leq \|A\|_2 \|x\|_2$,

$$\frac{\|A^{-1}f\|_2}{\|x\|_2} \leq \kappa_2(A)\frac{\|f\|_2}{\|b\|_2} \leq \frac{KnQ}{2S}.$$

Combining these estimates proves the explicit bound

$$\frac{\|e\|_2}{\|x\|_2} \leq \frac{3KnQ/(2S)}{1 - KnQ/S} \leq \frac{3}{2(16K-1)} \leq \frac{1}{10K}. \tag{8}$$

The last inequality is equivalent to $30K \leq 32K - 2$, valid for $K \geq 1$. This proves (1), including $n = 1$.

For the bit work, h and $s = \log_2 R$ are $O_c(\log(2n))$. Rounding and reading the input cost $O_c((m+n)\,\mathrm{polylog}(2n))$. Lemma 1 gives $\widetilde{O}_c(n(m+n))$ lifting and digit-assembly work. Lemma 2 gives $\widetilde{O}_c(n^2)$ reconstruction work. Multiplying the recovered numerators by R and writing the rational output also cost $O(n^2 s)$. The recovered denominators are odd, so this multiplication preserves reduced fractions without a further long-integer gcd. Since $m \geq n$, the total cost is $\widetilde{O}_c(nm)$. Every operation is deterministic. □

4 Implementation and finite checks

The accompanying Python library accepts a sparse list with zero-based indices and rational entries. Its default reconstruction backend calls FLINT's bounded rational reconstruction, through the pinned `python-flint 0.8.0` wheel [4]. The binding supports its 64-bit Linux and macOS ABI. Integer transfer uses hexadecimal strings, avoiding a decimal-conversion cost on long integers. Digits are stored in byte arrays and concatenated once. For this purpose the code rounds s up to a multiple of eight; the resulting R is at most 2^7 times the least admissible radix. Every bound above remains valid because $R \geq 2nT$ and $s = O_c(\log(2n))$ still hold. A separate classical Euclidean backend serves as a slow reference and carries no fast bit-complexity claim.

The tests compare the output with exact rational Gaussian elimination, exercise negative and zero coordinates, non-dyadic denominators, small input scales, and the condition-bound endpoint, and compare fast reconstruction with the Euclidean reference. A fixed 32,768-bit reconstruction test checks the long-integer interface. The reproduction script uses $c = 2$ and $n \in \{4, 8, 16, 32\}$ in each of four families: a diagonal matrix with entries $1, \ldots, 1, n^{-2}$; a scalar matrix $I/(2n^2 - 1)$; a tridiagonal matrix with alternating diagonal entries ± 2 and off-diagonal entries $1/4$; and an upper bidiagonal matrix with diagonal 1 and superdiagonal $1/4$. The right-hand sides have coordinates

$(-1)^i/(2i+1)$, indexed from zero. The first two families have condition numbers n^2 and 1. For the third, the off-diagonal operator has norm at most $1/2$, giving singular values in $[3/2, 5/2]$; for the fourth they lie in $[3/4, 5/4]$. Thus all cases meet the input promises.

All 14 tests and all 16 reproduced cases passed. Case acceptance uses the exact squared form of (1), with an independent rational solution as reference, together with the proved residual-width bound. Code, compact results, and reproduction instructions are available in the fixed version of the accompanying repository. These finite checks validate the implementation on the stated cases; the all-input guarantee and asymptotic work bound are established by the proof.

AI use. OpenAI Codex assisted with literature retrieval, derivation, proof checking, implementation, finite-case validation, and preparation of this manuscript. The central algorithm and the fast rational reconstruction result are attributed to Kelner. This statement does not assert independent human review or formal verification.

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